

Stem cell and bone marrow transplants

Stem cell and bone marrow transplants are sometimes used to treat types of cancer including leukaemia, lymphoma and myeloma.

On this page

- [What is a stem cell or bone marrow transplant?](#)
- [What is bone marrow?](#)
- [What are stem cells?](#)
- [Autologous stem cell transplants \(using your own cells\)](#)
- [Donor stem cell \(allogeneic\) transplants](#)
- [About our information](#)
- [How we can help](#)

What is a stem cell or bone marrow transplant?

Stem cell and bone marrow transplants are a type of treatment that is sometimes used to treat:

- some types of [leukaemia](#)
- some types of [lymphoma](#)
- [myeloma](#)
- some other blood conditions.

To understand how stem cell and bone marrow transplants work, it helps to know about the bone marrow and stem cells.

What is bone marrow?

Bone marrow is part of our immune system, which protects us from infection and disease. It is found inside our bones, mainly in the hip bone (pelvis) and the breastbone. The bone marrow is where stem cells are made.

Image: Bone marrow and blood cells

What are stem cells?

Stem cells are blood cells at the earliest stage of development. All our blood cells develop from stem cells in the bone marrow. When the blood cells are fully developed, they go into the bloodstream. Stem cells stay inside the bone marrow.

Blood cells do not live long. The bone marrow normally makes millions of new blood cells every day to replace blood cells as they are needed.

The three main types of blood cell are:

- red blood cells – these contain haemoglobin (Hb), which carries oxygen from the lungs to all the cells in our body
- platelets – these help the blood to clot and prevent bleeding and bruising
- white blood cells – these fight and prevent infection.

There are two main types of white blood cell. These are called neutrophils and lymphocytes. Neutrophils are the most common. You will hear your doctor or nurse talk about your neutrophil count during your treatment.

Autologous stem cell transplants (using your own cells)

An autologous stem cell transplant involves collecting and storing your own stem cells so they can be put back again after treatment. This means you can have much higher doses than usual of chemotherapy and sometimes radiotherapy to treat the cancer.

The high-dose treatment destroys the cancer cells, but it also affects your healthy blood cells. After the high-dose treatment, you are given your stem cells back through a drip (infusion). The stem cells travel through your blood to your bone marrow. They then begin to make new blood cells.

An autologous stem cell transplant is a complex treatment. This page gives some simple information, but you can read more on the [Anthony Nolan website](#).

Donor stem cell (allogeneic) transplants

A donor stem cell transplant aims to replace bone marrow that is no longer working properly with healthy stem cells from another person (a donor). This type of transplant is also called an allogeneic stem cell transplant.

You have chemotherapy and sometimes other treatments to:

- get your body ready to accept the donor stem cells
- weaken your immune system to try and stop it attacking the donor stem cells
- remove any remaining cancer cells from your body.

You are then given the donor's stem cells through a drip (infusion). A donor stem cell transplant is a complex treatment. This page gives some simple information, but you can read more on the [Anthony Nolan website](#) including information about side effects such as [graft versus host disease](#).

About our information

References



Below is a sample of the sources used in our stem cell and bone marrow transplants information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Carreras, Dunfour, Mohty and Kröger. The European Society for Blood and Bone Marrow Transplantation: Hematopoietic stem cell transplantation and cellular therapies. Springer, Switzerland. 2019.

FACT-JACIE International Standards for Hematopoietic Cellular Therapy – product collection, processing and administration. 7th edition. 2018.

Anthony Nolan. Operations and patient services user guide. Available from www.anthonynolan.org. Accessed January 2019.

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

-

use plain English

- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 December 2025 | Next review: 01 December 2028

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help